```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("Unemployment in India.csv")
```

```
df.head()
```

	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Area
0	Andhra Pradesh	31-05-2019	Monthly	3.65	11999139.0	43.24	Rural
1	Andhra Pradesh	30-06-2019	Monthly	3.05	11755881.0	42.05	Rural
2	Andhra Pradesh	31-07-2019	Monthly	3.75	12086707.0	43.50	Rural

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 7 columns):
 #   Column                                    Non-Null Count  Dtype  
---  ------                                    --------------  -----  
 0   Region                                    740 non-null    object 
 1    Date                                     740 non-null    object 
 2    Frequency                                740 non-null    object 
 3    Estimated Unemployment Rate (%)          740 non-null    float64
 4    Estimated Employed                       740 non-null    float64
 5    Estimated Labour Participation Rate (%)  740 non-null    float64
 6   Area                                      740 non-null    object 
dtypes: float64(3), object(4)
memory usage: 42.1+ KB
```

```
df.describe()
```

	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)
count	740.000000	7.400000e+02	740.000000
mean	11.787946	7.204460e+06	42.630122
std	10.721298	8.087988e+06	8.111094
min	0.000000	4.942000e+04	13.330000
25%	4.657500	1.190404e+06	38.062500
50%	8.350000	4.744178e+06	41.160000
75%	15.887500	1.127549e+07	45.505000
max	76.740000	4.577751e+07	72.570000

```
df.columns
```

```
Index(['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate (%)',
       ' Estimated Employed', ' Estimated Labour Participation Rate (%)',
       'Area'],
      dtype='object')
```

```
df.isnull().sum()
```

	0
Region	28
Date	28
Frequency	28
Estimated Unemployment Rate (%)	28
Estimated Employed	28
Estimated Labour Participation Rate (%)	28
Area	28

dtype: int64

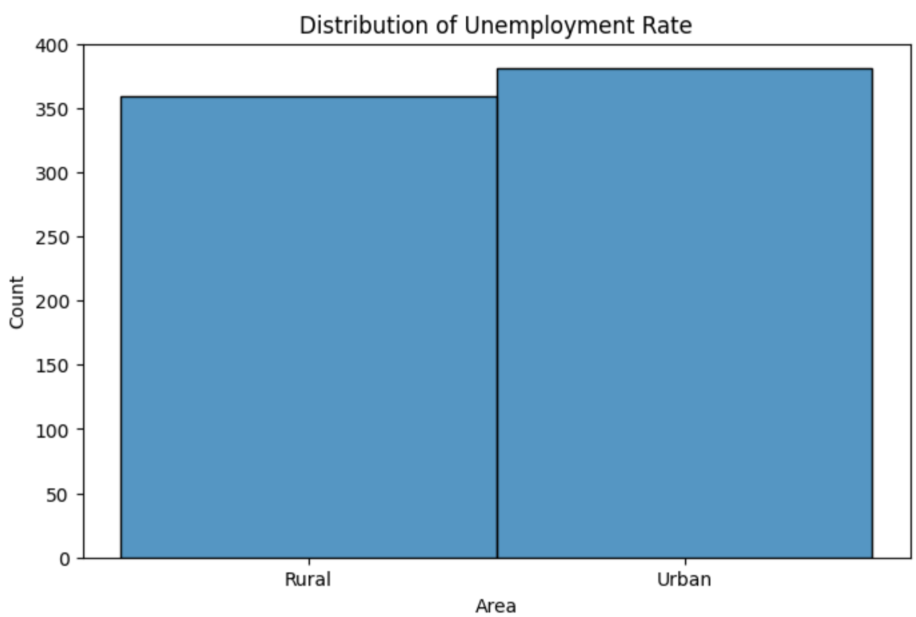
```
df.duplicated().sum()
```

```
np.int64(27)
```

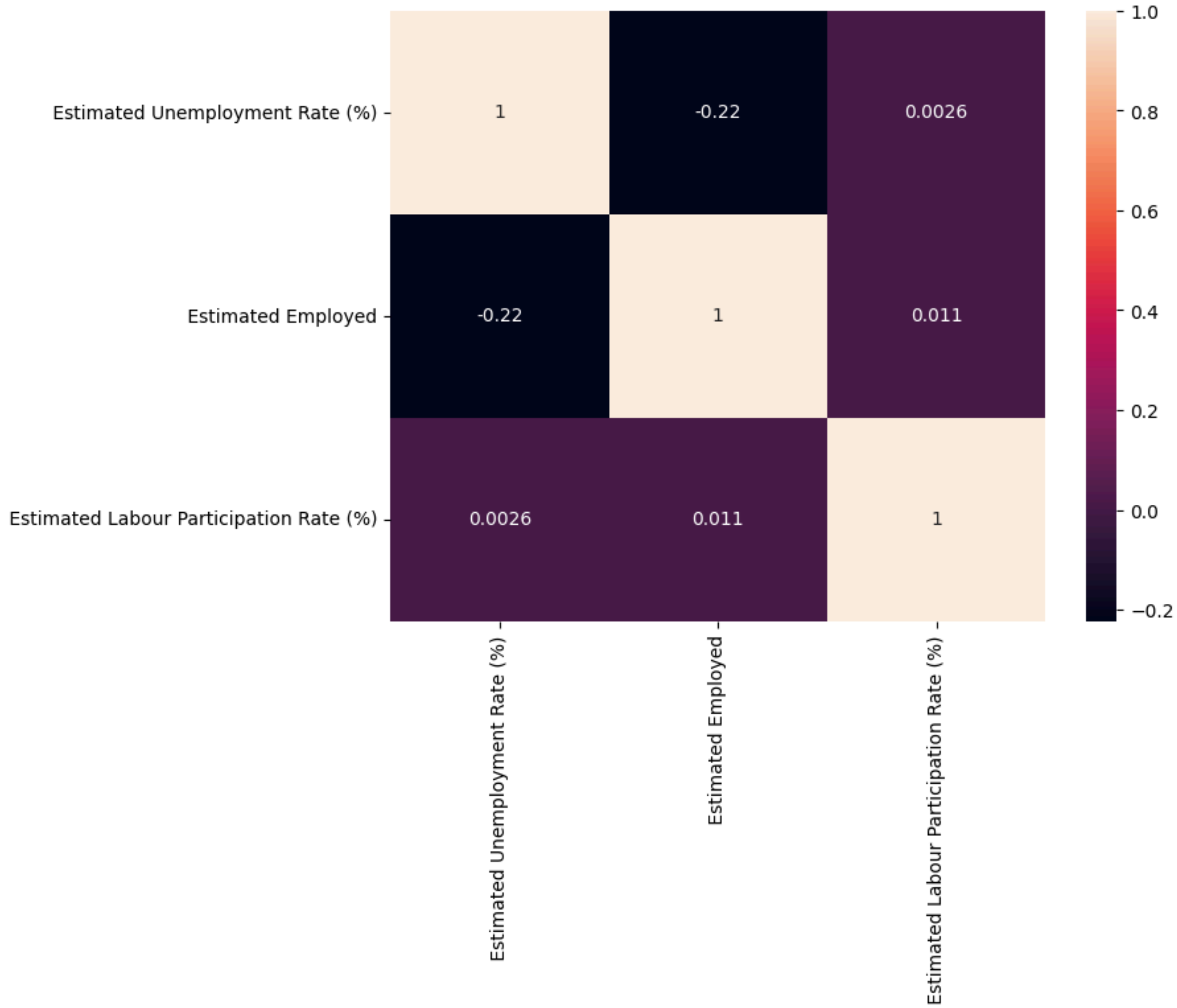
```
df.drop_duplicates(inplace=True)
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
plt.figure(figsize=(8,5))
sns.histplot(df.iloc[:, -1], bins=20)
plt.title("Distribution of Unemployment Rate")
plt.show()
```




```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```




```
df.isnull().sum()
```

	0
Region	1
Date	1
Frequency	1
Estimated Unemployment Rate (%)	0
Estimated Employed	0
Estimated Labour Participation Rate (%)	0
Area	1

dtype: int64

```
df.isnull().sum()
```

	0
Region	1
Date	1
Frequency	1
Estimated Unemployment Rate (%)	0
Estimated Employed	0
Estimated Labour Participation Rate (%)	0
Area	1

dtype: int64

```
df = df.dropna()
```

```
df.isnull().sum()
```

	0
Region	0
Date	0
Frequency	0
Estimated Unemployment Rate (%)	0
Estimated Employed	0
Estimated Labour Participation Rate (%)	0
Area	0

dtype: int64

```
df.columns = df.columns.str.strip()
print(df.columns)
```

```
Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
       'Estimated Employed', 'Estimated Labour Participation Rate (%)',
       'Area'],
      dtype='object')
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
/tmp/ipykernel_711/2394721818.py:1: UserWarning: Parsing dates in  %d-%m-%Y format when dayfirst=False (the default) was specifi
  df['Date'] = pd.to_datetime(df['Date'])
```

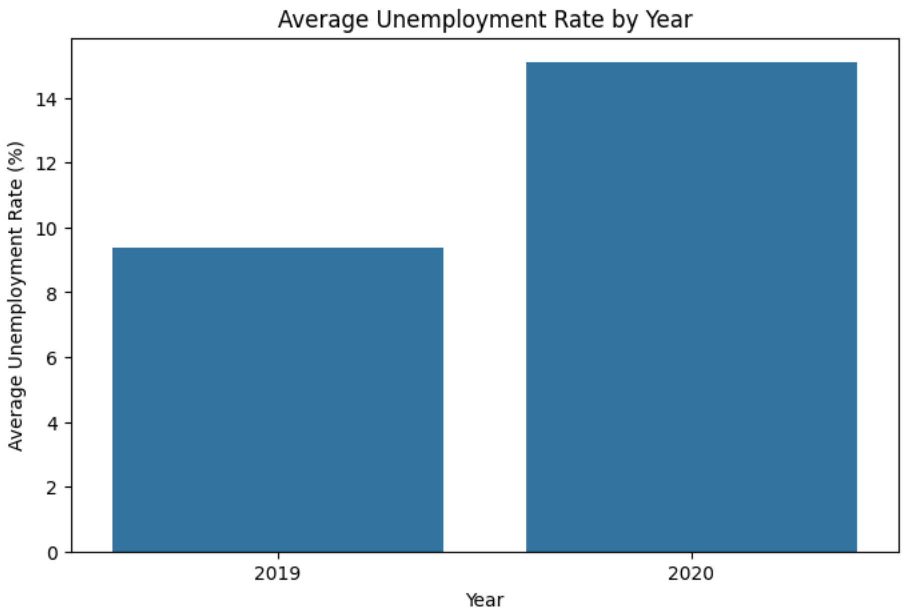
```
df['Year'] = df['Date'].dt.year
```

```
df.groupby('Year')['Estimated Unemployment Rate (%)'].mean()
```

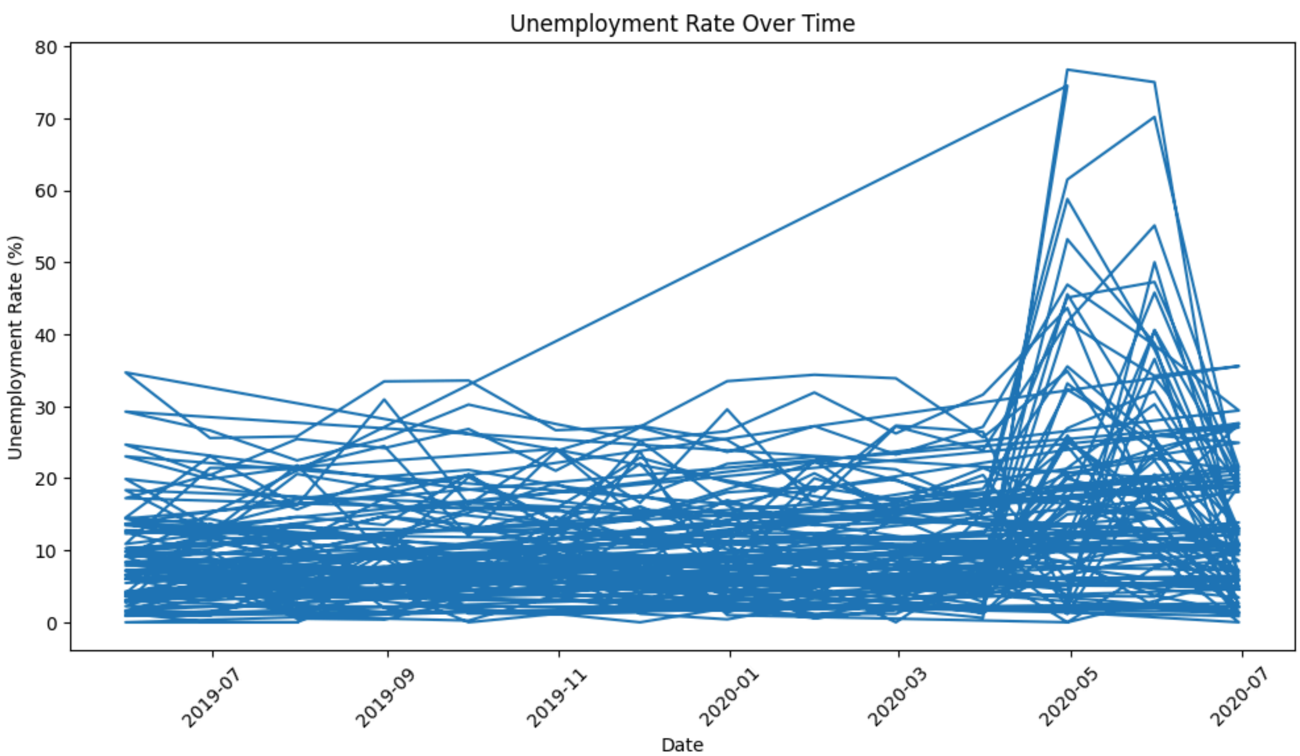
Year	Estimated Unemployment Rate (%)
2019	9.399047
2020	15.101581

dtype: float64

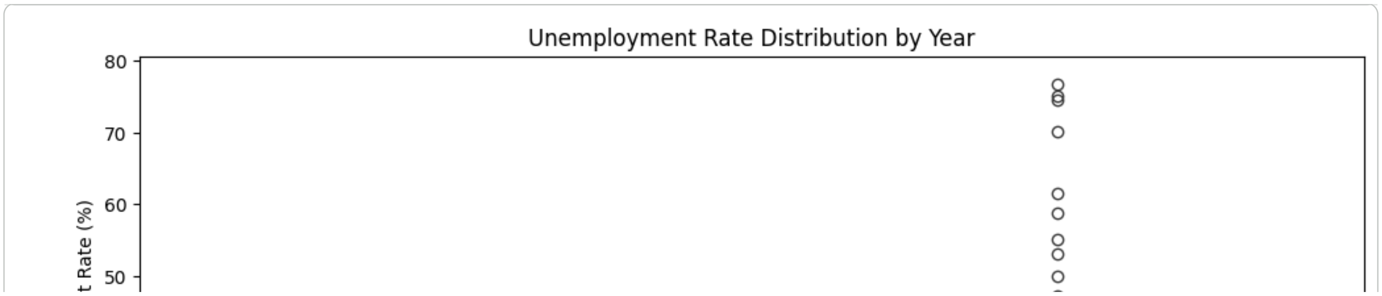
```
plt.figure(figsize=(8,5))
sns.barplot(
    x=yearly_unemployment.index,
    y=yearly_unemployment.values
)
plt.title('Average Unemployment Rate by Year')
plt.xlabel('Year')
plt.ylabel('Average Unemployment Rate (%)')
plt.show()
```




```
plt.figure(figsize=(12,6))
plt.plot(df['Date'],
         df['Estimated Unemployment Rate (%)'])
plt.title("Unemployment Rate Over Time")
plt.xlabel("Date")
plt.ylabel("Unemployment Rate (%)")
plt.xticks(rotation=45)
plt.show()
```




```
plt.figure(figsize=(12,6))
sns.boxplot(
    data=df,
    x='Year',
    y='Estimated Unemployment Rate (%)'
)
plt.title('Unemployment Rate Distribution by Year')
plt.show()
```


Unemployment Rate Distribution by Year
80
70
60
50
t Rate (%)